Evaluating Energy Saving Techniques in Networks

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***Abstract*—Energy consumption of internet connected devices is growing exponentially. This report investigates methods of reducing consumption on a simulated wired network. A network of 11 routers was simulated across three scenarios using OSPF routing protocol. Scenario 1 provided a baseline of all routers awake. Scenario 2 placed only routers that saw no traffic to in scenario 1. Scenario 3 slept the central routers of the topology meaning traffic was diverted around different routes. Scenario 2 saw a 0.32% reduction in energy consumption, scenario 3 saw a 28% increase in energy consumption. These results demonstrated smart sleeping techniques are viable, but node selection is important.**

Key words: *Energy Efficiency, Smart Sleeping, OSPF, Wired Networks*

I.Introduction

As the number of devices connected to the internet each person possesses increases, so does the energy burden of the internet. The ICT sector is estimated to account for approximately 8% of worldwide energy consumption with network equipment contributing a significant portion of that figure (Cianfrani et al., 2012). As the demand for networking equipment and higher bandwidth continues to grow each year so does the importance of reducing energy wastage. Currently, ethernet is the dominant technology for LANs valued in 2020 at $10.49 billion USD (Ashokkumar, 2025). This means that small reductions in energy consumption can contribute to a substantial global saving (Christensen et al., 2010).

This report aims to test energy usage on a network and examine what methods are possible to reduce energy usage when transmitting data.

II. Estimating Energy Consumption in wired networks

Many studies on estimating energy consumption have been done, this report will cite (Texas Instruments, 2013). This report is data that was collected with the aim to provide accurate power consumption in typical Networks. Therefore, the paper defines, in a typical 100 Base-Tx Operating Mode, the energy consumption for a packet is 294.65 Milliwatt (mW), this paper will round this up to 295 mW per packet. Following this, this paper will define that a control packet will consume approximately half the energy as a packet 148 mW.

(Texas Instruments, 2013) also defines the energy usage of idle routers as 285.81 Mw, rounded to 286. Along with the energy usage for a powered down or sleeping router as 24.22, rounded to 24 mW.

A. Smart Sleeping

A proposed solution to the problem of increasing energy usage is a method named smart sleeping, this idea consists of sending networked routers, that are not actively sending or receiving a signal, to go to sleep, with the occasional refresh signal to maintain alignment in the network. (Christensen et al., 2010) explains the IEEE 802.3az, a standard approved in 2010, could save over approximately $1 billion USD worldwide. More recent papers claim that a 70% reduction in energy usage can be achieved by incorporating the low-power idle mode. (Ashokkumar, 2025)

B. Energy Efficient Routing Protocols

To create greater efficiency in networks, various energy efficient routing strategies have been explored. One approach involves modifying existing routing protocols such as Open Shortest Path First (OSPF). This method aims to consolidate traffic on fewer active paths during low traffic periods. This approach can be beneficial in ISP backbones, as these networks are often not fully utilised outside of peak traffic times (Cianfrani et al., 2012).

Energy Saving IP Routing (ESIR) is a solution that can be integrated with OSPF so that the usual effects of IP topology reconfiguration is avoided. Utilising ESIR, a network can see a reduction of 30% of its network links with minimal impact on network performance, greatly cutting down on energy usage (Cianfrani et al., 2012).

III. Critical evaluation

This experiment consisted of a network with 11 Routers on an OSPF network, loosely inspired by GÉANT, with 4 wired nodes on the four corners of the network. Two of the nodes sent and received Constant Bit Rate (CBR) packets to each other through the network. Two different nodes were sending and receiving email packets to each other. The total network was then simulated for 420 milliseconds (ms) consisting of 30 ms intervals. The experiment ran three times, first the control where all routers were awake and in a ready state, followed by two more scenarios consisting of different sleep patterns between the routers.

Scenario 1 consists of running the experiment while all nodes are awake and in an idle position. This meant that the routers that saw no traffic for the whole simulation, however still used energy, this scenario is expected to have the highest energy consumption.

Following scenario 2 took all the routers that in scenario 1 saw no traffic and put them to sleep after the first 30ms interval, this meant that only the routers that needed to be awake to transfer data around the network were awake. This assumed that since these routers saw no traffic to begin with, they were not needed to transmit the packets within the simulation. Because of this drop in routers using energy, this scenario saw a reduced energy consumption of 0.32% compared to scenario 1. The energy saving is small due to the only difference being they are now asleep rather than idling like in the previous simulation.

Finally in scenario 3, the central nodes of routers 4 and 10 were put to sleep, meaning that any traffic had to be routed around the outside of the network. This scenario had the assumption that rather than one router in the centre handling the most traffic, load would be shared across the outsides of the network. This however had the opposite effect, since traffic could no longer take the optimal route, there was a large energy increase of around 28% when compared to scenario 1, this is because packets had a longer journey time in comparison to the previous scenarios.

These finding are consistent with the literature reviewed in section II. The reduction observed in scenario 2 aligned with the principles outlined by (Christensen et al., 2010). Whereby placing idle network devices in a sleep state can result in a reduction of energy. However as highlighted in (Cianfrani et al., 2012) the effectiveness of sleep strategy is entirely dependent on which nodes are selected as their ESIR approach specifically avoids sleeping central routing nodes. This can be seen demonstrated in scenario 3 where the network saw a 28% increase in energy consumption. This suggests that in a real-world network, a more sophisticated approach such as ESIR would be necessary to maximise energy savings while avoiding the shortcomings observed in Scenario 3.

Figures can be seen below:

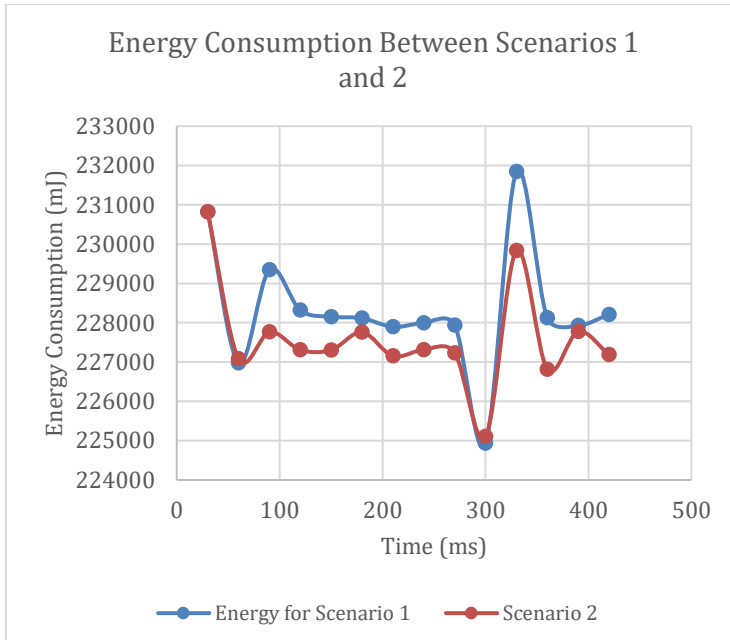

Figure 1: The energy consumption of Scenarios 1 and 2.

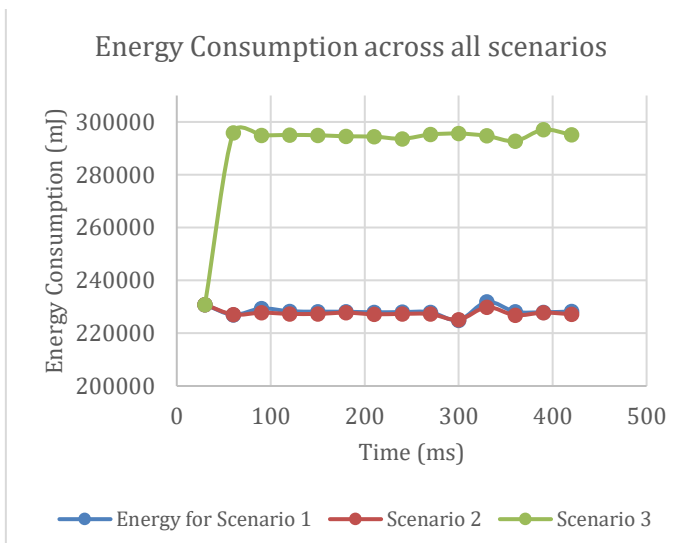

Figure 2: The energy consumption of Scenarios 1, 2, and 3.

IV.Conclusions

To conclude, this report evaluated the energy consumption of a network in three separate scenarios. Using scenario 1 as a baseline the subsequent scenarios tested methods that could be used to reduce wasted energy. Scenario 2 found that just by sending routers that saw no traffic to sleep, a small but measurable reduction in energy consumption could be seen. However, removing the central routers of a network forces packets to find an alternate, and likely longer path which greatly increased the energy usage seen in scenario 3.

These results demonstrate that smart sleeping is a viable method of reducing energy consumption, however node selection is critical so that the maximum reduction can be achieved.

Future work could explore implementing a more sophisticated sleeping strategy, such as ESIR where dynamic identification of nodes can be used to sleep unused nodes.

References

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